Computer Vision Lab Mid Exam

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Registration: FA23-BAI-018

Topic: HOG vs SIFT on BloodMNIST

Q1 - Dataset Setup

Training shape: (400, 128, 128)

Validation shape: (120, 128, 128)

Test shape: (120, 128, 128)

Image size: 128 x 128 grayscale

Classes: Eosinophil, Monocyte, Neutrophil, Platelet

Subset: 100 training, 30 validation, 30 test images per class

Q2 - HOG + SVM

Cells per dimension: 128 / 8 = 16

Blocks per dimension: 16 - 2 + 1 = 15

Values per block: 2 x 2 x 9 = 36

Total HOG features: 15 x 15 x 36 = 8100

Selected C: 0.1

Validation accuracy: 0.7167

Validation macro-F1: 0.7231

HOG validation confusion matrix:

[[20 7 3 0]

 [8 21 1 0]

 [4 8 18 0]

 [1 2 0 27]]

Q3 - SIFT + SVM

Average normal SIFT keypoints: 71.92

Minimum keypoints: 10

Maximum keypoints: 187

Zero-keypoint images: 0

Fixed grid: 6 x 6 = 36 keypoints

Keypoint size: 32

Descriptor length: 128

Feature length: 36 x 128 = 4608

Selected C: 0.1

Validation accuracy: 0.7667

Validation macro-F1: 0.7678

SIFT validation confusion matrix:

[[17 11 2 0]

 [7 22 0 1]

 [3 2 24 1]

 [0 1 0 29]]

Q4 - Robustness Results

Model	Condition	Accuracy	Macro-F1	Precision (macro)	Recall (macro)
HOG + SVM	Clean	0.7083	0.7113	0.7151	0.7083
HOG + SVM	Rotated 20 degrees	0.7250	0.7281	0.7387	0.7250
HOG + SVM	Gaussian noise sigma=0.05	0.2500	0.1000	0.0625	0.2500
SIFT + SVM	Clean	0.6833	0.6805	0.6865	0.6833
SIFT + SVM	Rotated 20 degrees	0.6833	0.6856	0.6924	0.6833
SIFT + SVM	Gaussian noise sigma=0.05	0.6917	0.6844	0.6840	0.6917

MODEL COMPARISON

Model	Selected C	Clean Accuracy	Clean Macro-F1	Rotated 20 degrees Accuracy	Rotated 20 degrees Macro-F1	Gaussian noise sigma=0.05 Accuracy	Gaussian noise sigma=0.05 Macro-F1
HOG + SVM	0.1	0.7083	0.7113	0.7250	0.7281	0.2500	0.1000
SIFT + SVM	0.1	0.6833	0.6805	0.6833	0.6856	0.6917	0.6844

Q4 - Class-Level Metrics

Model	Class	TP	TN	FP	FN	Precision	Recall	Specificity
HOG + SVM	Eosinophil	15	75	15	15	0.5000	0.5000	0.8333
HOG + SVM	Monocyte	19	77	13	11	0.5938	0.6333	0.8556
HOG + SVM	Neutrophil	23	83	7	7	0.7667	0.7667	0.9222
HOG + SVM	Platelet	28	90	0	2	1.0000	0.9333	1.0000
SIFT + SVM	Eosinophil	16	72	18	14	0.4706	0.5333	0.8000
SIFT + SVM	Monocyte	15	83	7	15	0.6818	0.5000	0.9222
SIFT + SVM	Neutrophil	21	79	11	9	0.6562	0.7000	0.8778
SIFT + SVM	Platelet	30	88	2	0	0.9375	1.0000	0.9778

Q5 - Answers and Formulas

1. HOG block normalization:
It scales local gradient histograms to reduce sensitivity to illumination and contrast changes. L2-Hys limits large values.

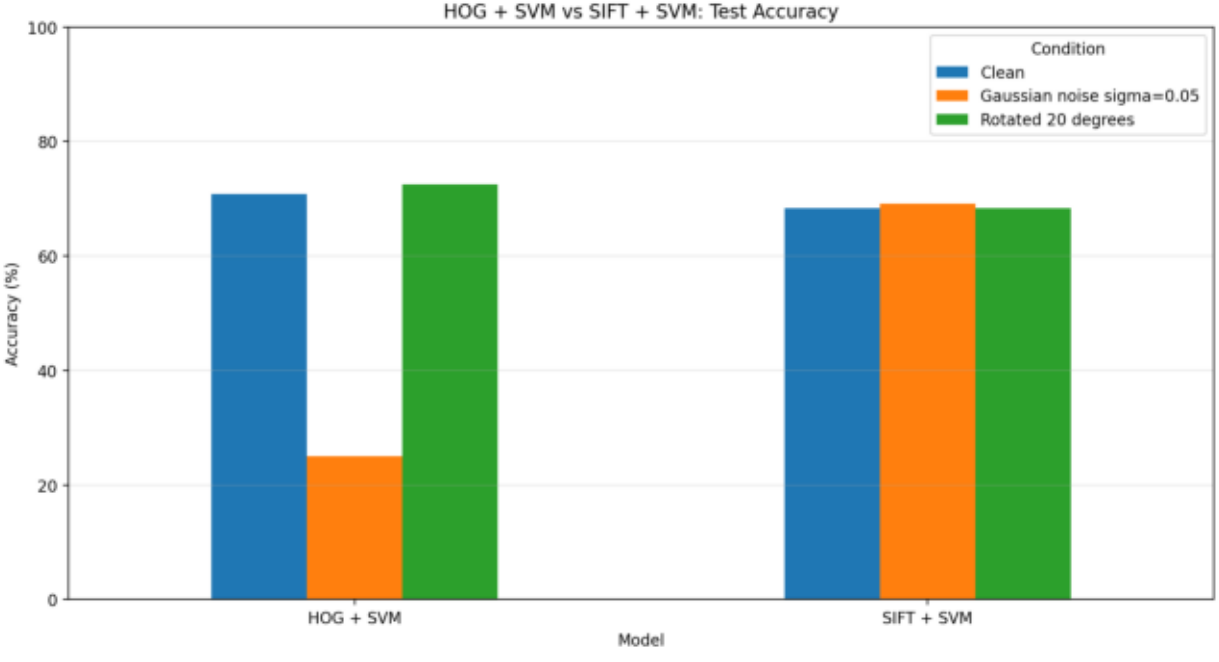
2. Why not choose C on the test set?
That leaks final evaluation information into model selection. Use validation data to select C and reserve test data for final evaluation.

3. Most confused classes:
Use the largest off-diagonal entry in each clean-test confusion matrix. Grayscale removes colour information and may make similar classes harder to distinguish.

4. AI log:
Prompt: Build Q1-Q5 HOG/SIFT lab notebook.
Change: Registration-specific settings and fixed-grid SIFT.
Check: Feature lengths, plots, metrics and confusion matrices.

FORMULAS
Accuracy = (TP+TN)/(TP+TN+FP+FN)
Precision = TP/(TP+FP)
Recall = TP/(TP+FN)
Specificity = TN/(TN+FP)
F1 = 2*Precision*Recall/(Precision+Recall)
Macro-F1 = mean of per-class F1 scores.

accuracy comparison



macro f1 comparison

